CMAI 480

Computer Vision

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PROJECTION GEOMETRY

WORLD 2D: PLANER PROJECTIVE TRANSFORMATIONS AND TRANSFORMATION GROUPS

Points, Lines and Conics
Homogeneous Coordinates

- From now on, we will call on algebraic structure homogeneous if it conveys real-world information through ratios.
- Homogeneous structures you have seen so far:
  1. 3-vectors $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ to represent points (x, y) in $\mathbb{R}^2$.
  2. 3-vectors $l = \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix}$ to represent lines in $\mathbb{R}^2$.
  3. 3×3 matrices $C = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{12} & c_{22} & c_{23} \\ c_{13} & c_{23} & c_{33} \end{bmatrix}$ to represent conics in $\mathbb{R}^2$.

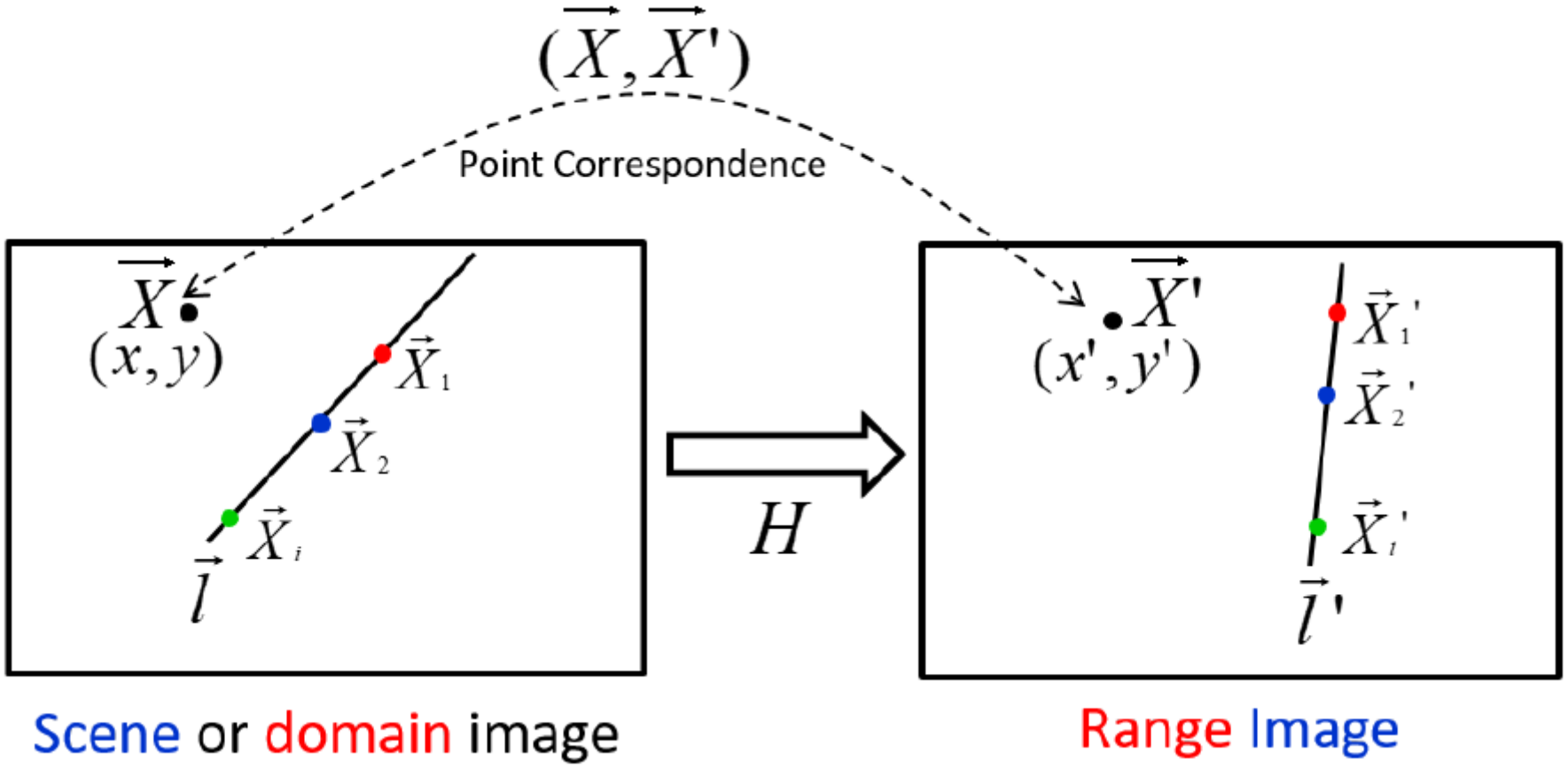
Planer Projective Transformation

- Let's now talk about constructing mappings from one (x, y)-plane to another (x, y)-plane.
- If we denote a mapping by H, we are obviously talking about $H : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.
- Representing the points in the planes by homogeneous 3-vectors, we have $H : \mathbb{R}^3 \rightarrow \mathbb{R}^3$.
- A Planer Projective Transformation is a linear transformation on homogeneous 3-vectors, the transformation being represented by a $\underbrace{non-singular}_{\text{Therefore } H^{-1} \text{ always exists}}$ 3×3 matrix H (full rank), as in:
  - $\begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix} = H \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Longrightarrow X' = HX$

Planer Projective Transformation

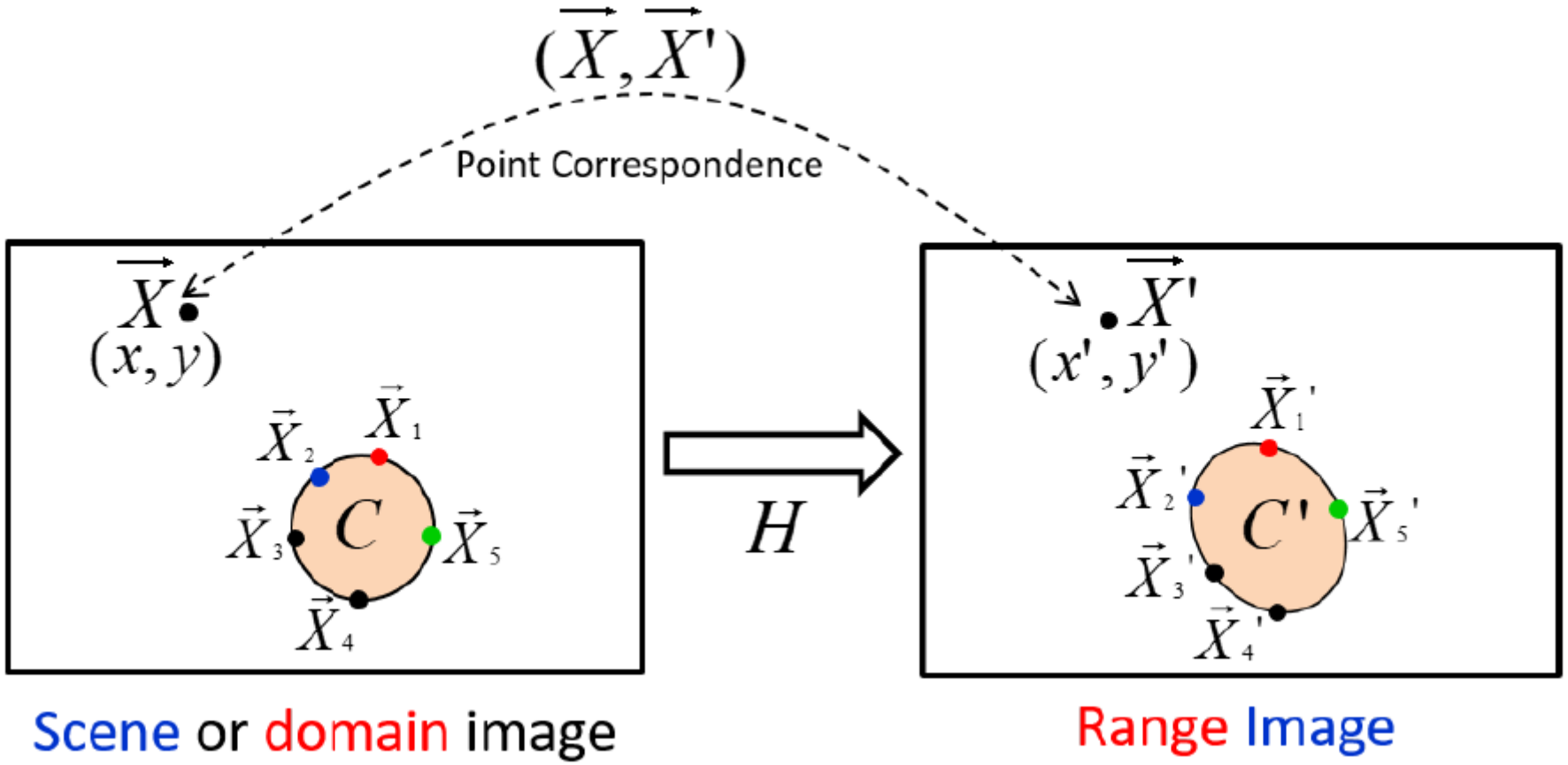
- Since only the ratios of the 3-vectors are important to us, it is only the ratios of the elements of H that matter. In other words, H multiplied by an arbitrary non-zero scalar does not alter the transformation. So we say H is homogeneous.
- **A planer projective transformation is more popularly known as Homography.**

A fundamental property of a homography is that it always maps a straight line to a straight line



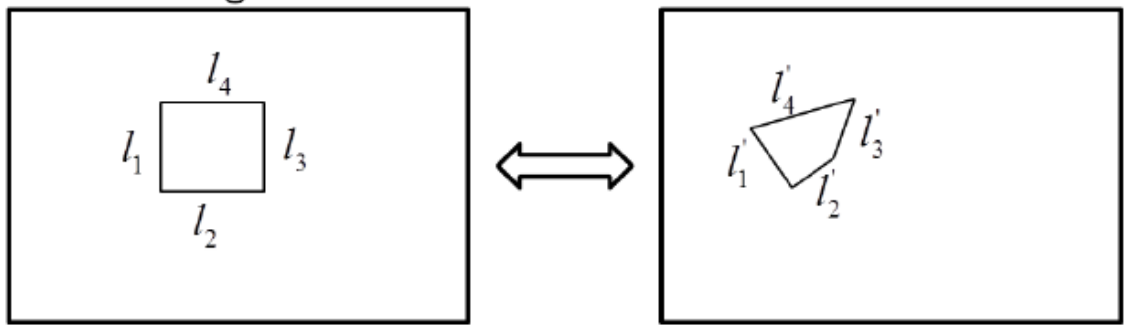
- Points transform according to H, $\quad \vec{X'} = H\vec{X}$
- Lines transform according to H^{-T}. $\quad \vec{l'} = H^{-T}\vec{l}$

A Conic C is Mapped According to $C' = H^{-T} C H^{-1}$



Planer Homography: Possible Visual Distortions

- Considering that straight lines go into straight lines, what sort of visual distortion might we expect from a general homography?
  - You could see a scene such as the one shown at left below turns into a scene that is shown at right:



Scene Image

  - This is what you would get if you took a photograph of a windows and picture frames mounted on a wall when the objects you are photographing are high on the wall and off to a side in relation to where you are standing.
    - The amount of projective distortion that one would get depends on how the camera is oblique with respect to the wall (Oblique Angle).

Removing Projective Distortion

- Will come to this topic very soon!!!







What distortion effects you can see?

Hierarchy of Transformations:
Projective Linear PL(3) is at the root of the hierarchy

- The specific subgroups of $PL(3)$ that we are interested in are the Affine, the Similarity, and the Euclidean, as shown in the figure at below:

Projective Groups PL(3)
↑ *subgroup of*
Affine Group
↑ *subgroup of*
Similarity Group
↑ *subgroup of*
Euclidean Group

Hierarchy of Transformations:

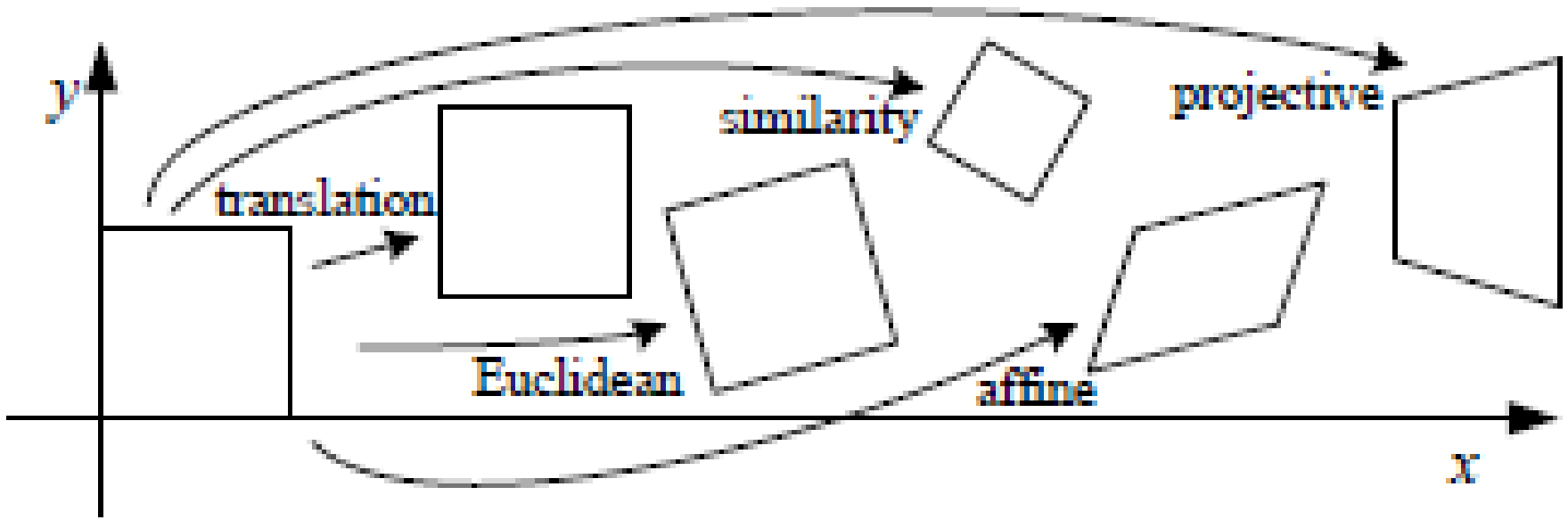
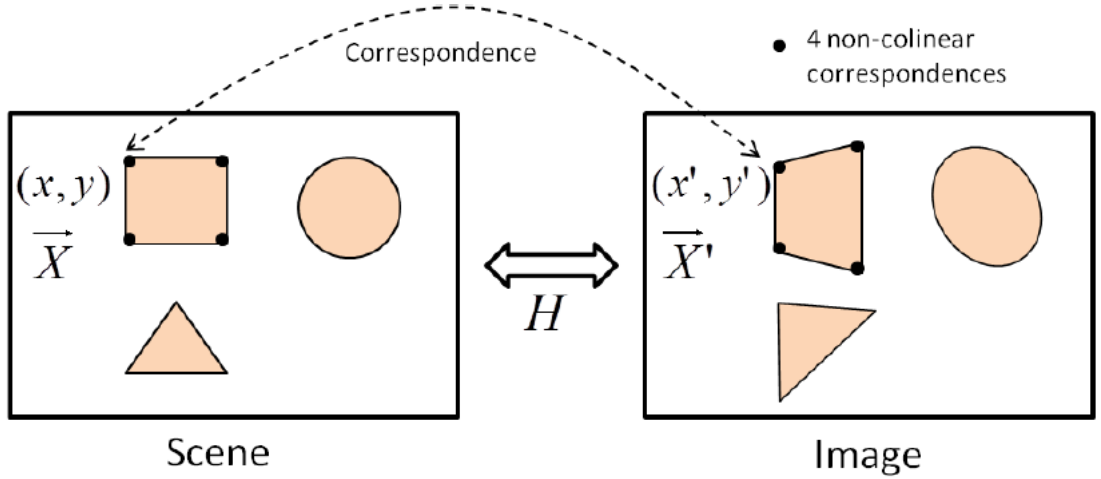


Figure 2.4 Basic set of 2D planar transformations.

Projective Linear Group PL(3)



Two Fundamental Questions:

1. What is the geometry invariant under the given PL transformation.
   i.e. What stays invariant under the given PL mapping $\mathbb{R}^2 \rightarrow \mathbb{R}^2$
2. What is the number of DOF (degree of freedom) of the given mapping.

- $\vec{X}' = H\vec{X}$
- $$\begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

The Engineering Benefits of Recognizing the Group Structures

- What are the engineering benefits of recognizing the group structures shown above?
  - The group structures guarantee us that, say, two successive applications of the affine transform (with two different matrices) will be affine.
  - For another example, applying first an affine transform followed by a similarity transform will result in an overall application of an affine transform.
    - Because a similarity transform is an affine transform.

The General Planer Transform H

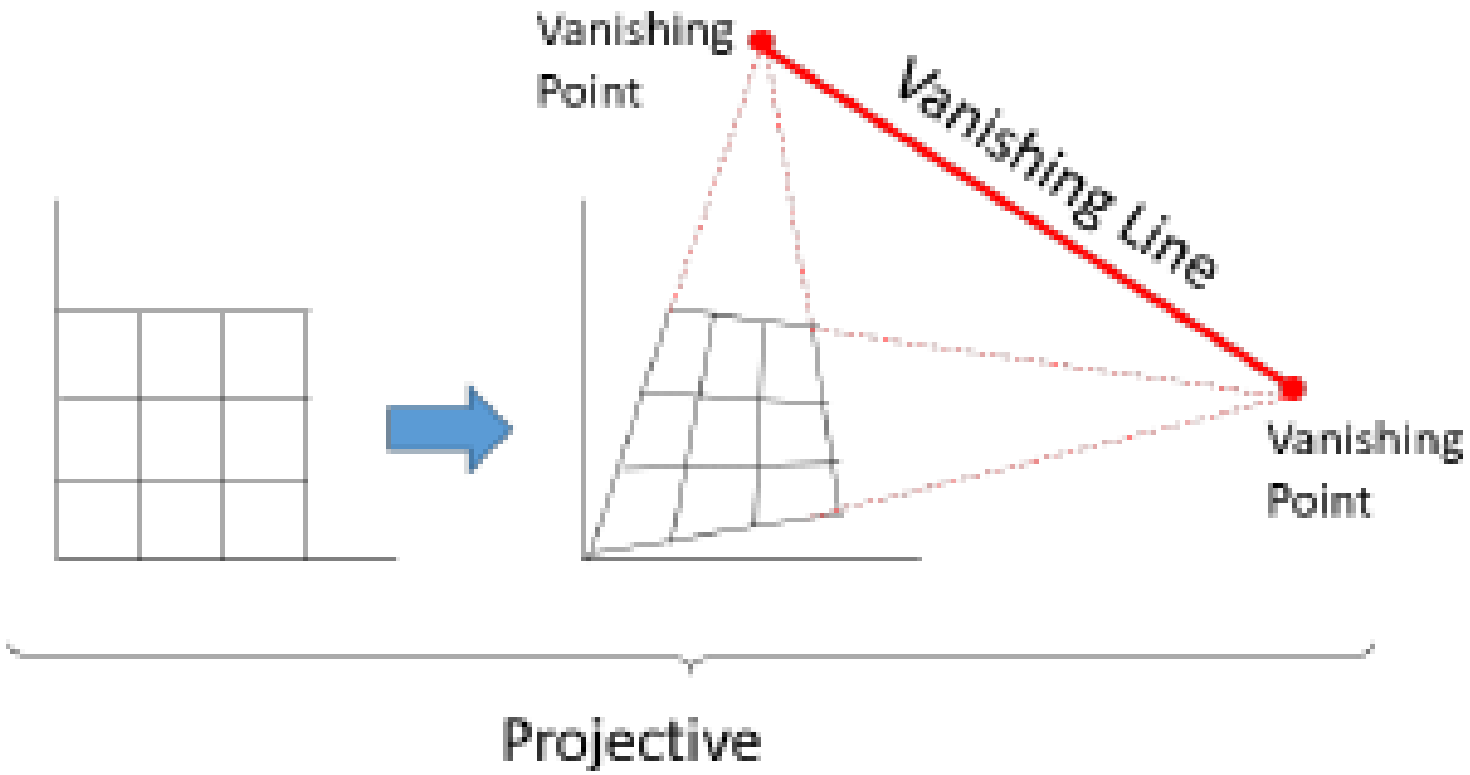
- Before providing definitions for the three subgroups of PL(3), let's first express the general planer transform in the following manner:

$$H = \begin{bmatrix} \mathrm{A} & \vec{t} \\ \vec{v}^T & v \end{bmatrix}_{3\times 3} \text{ with } \mathrm{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \vec{t} = \begin{pmatrix} t_x \\ t_y \end{pmatrix}, \vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

  The reason for the choice of symbols will become clear shortly.

- Since H is homogeneous — meaning that only the ratios of the elements of H are important — we will frequently set the element $v = 1$.
  - [However, beware, there will be situations when we will have to allow v to become zero. More on that later.]
- DOF: is 8
- We are now ready to define the three subgroups of PL(3).

Projective Transformations



Affine Transformations

- The affine group is obtained from the projective group by restricting the last row of the 3×3 transformational matrix to $\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$. You can easily show yourself that if we multiply two 3×3 matrices, each with its last row restricted to $\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$, the result is a matrix with its last row as $\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$.
- Affine transformations are commonly expressed as:

$$\begin{pmatrix} x_1' \\ x_2' \\ 1 \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix}$$

(x_1, x_2) and $\left(x_1', x_2'\right)$ are physical coordinates in their respective planes.

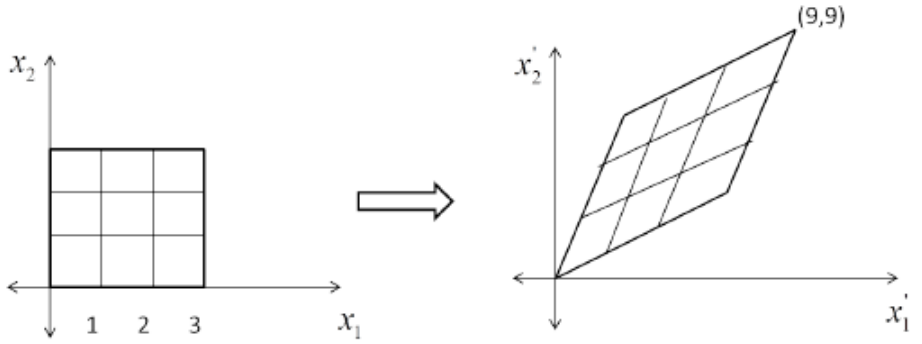
$\mathrm{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is non-singular ($i.e.$ $\det(\mathrm{A}) \neq 0$)

Affine Transformations

- To better understand the roles played by the different elements of an affine transformation matrix, we will first consider the case when $t_x = t_y = 0$.

Affine Transformations

- With t_x and t_y set to zero, consider an example of affine transformation:
  $X_1' = 2x_1 + x_2$
  $X_2' = x_1 + 2x_2$
  $\Longrightarrow A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:
- Using the above mapping, the following points are mapped as follows $(1,1) \longrightarrow (3,3)$ and $(3,3) \longrightarrow (9,9)$.



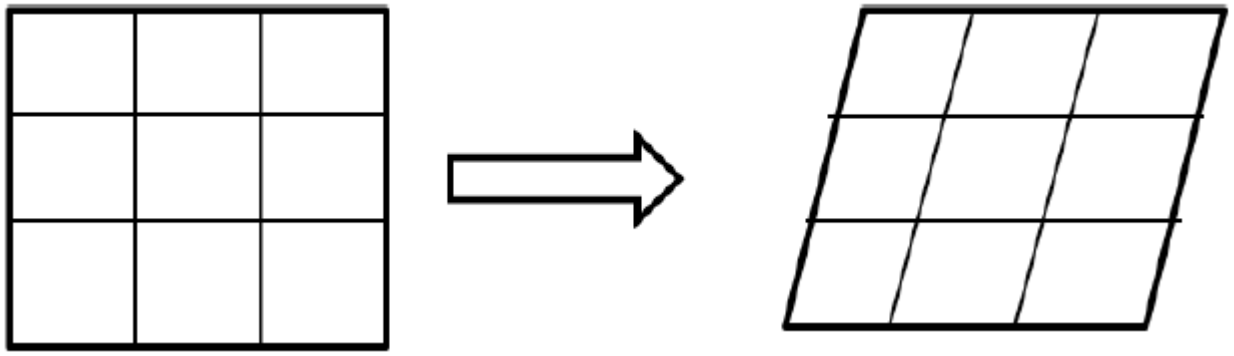
- As you can see, an affine transformation not only takes straight lines into straight lines, it also keeps parallel lines parallel. While stretching the plane in one direction, it shrinks it in the orthogonal direction, but in such a way that parallel lines stay parallel.

Classic Affine Distortion Effect with t_x and t_y set to zero

- Differential Effect
  - One side compressed and the other one stretched
- Rotation + compression + dilation (expansion).
- All the above effects are caused by the non-singular 2×2 A matrix.

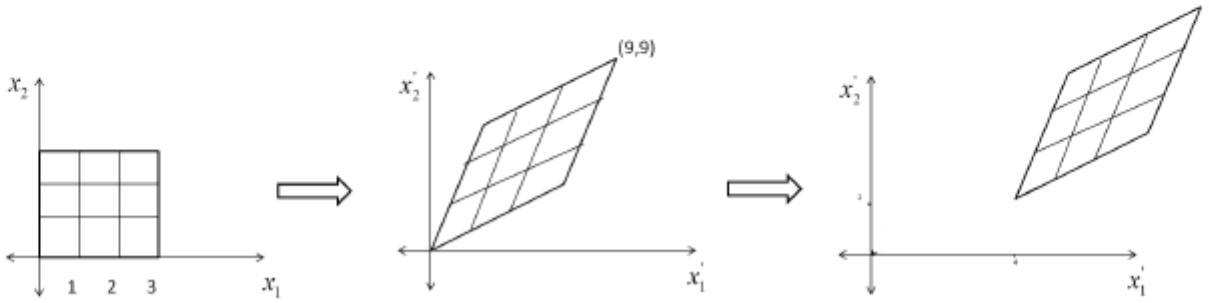
Affine Transformation

- What stays invariant in affine group?
  - 4 or 5 properties are mentioned in the book
    - Example: area
  - We will focus only on one important invariant
    - Parallel lines stay parallel



Classic Affine Distortion Effect with t_x and t_y set to non-zero

- With regard to the role played by t_x and t_y, it is easy to show that these simply translate the pattern after it has been rotated and scaled by A.
- Just try using $t_x = 4$ and $t_y = 2$ in the example we have seen before.

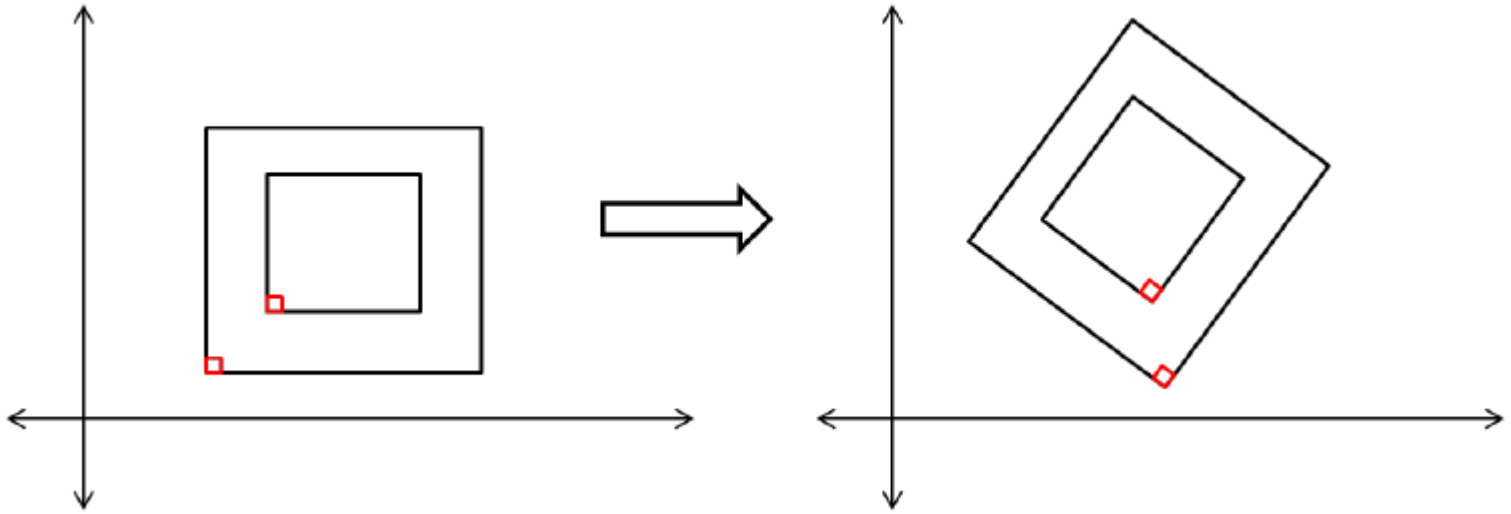


Similarity Transformations

- The similarity group is obtained from the affine group by requiring that A be orthogonal: $A^T A = \lambda^2 \mathrm{I}$.
  - 4DOF
- Similarity group effects:
  - Volume expansion or contraction and reflection
- **Similarity transform preserves angles**.
  - Physical relation between physical coordinates stay preserved.

Similarity Transformations

- Shape and angles are preserved
  - Also: area
- But, for the distances they are not preserved
  - The distances either extended or shrinked based on λ.



Similarity Transformations

- Try: $\begin{pmatrix} x_1' \\ x_2' \\ 1 \end{pmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix}$

Euclidean Transformations

- The euclidean group is obtained from the similarity group by further requiring that A be orthonormal.
- That is $A^T A = I$
  - I is a 2×2 identity matrix
- This group represents transformations corresponding to rigid body motions.
- Preserves Euclidean distance between any two points in the domain plane.
  - i.e. distance is invariant under the the euclidean group

Review

- Projective group:
  - Straight line invariant
  - 8DOF
- Affine group
  - Parallel lines stay parallel
  - The only condition is that the A matrix is non-singular.
  - 6DOF
- Similarity group
  - The A matrix must be orthogonal
  - 4DOF
- Euclidean group
  - The A matrix must be orthonormal
  - 3DOF

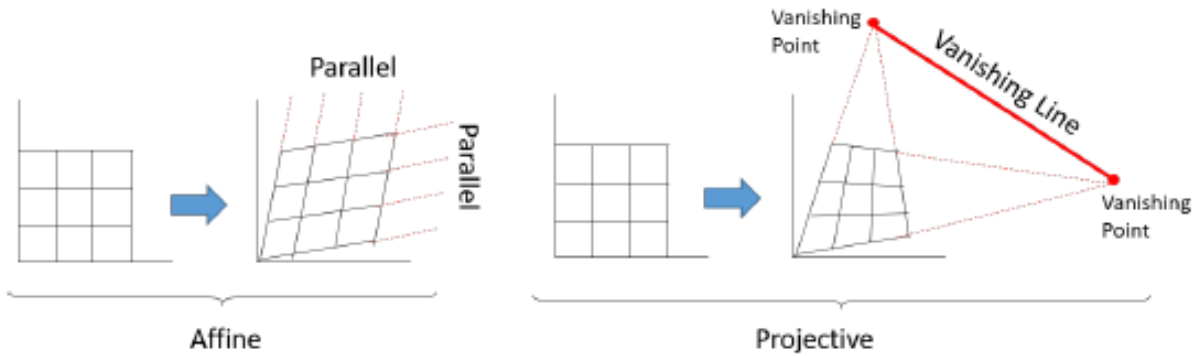
Review

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\left[\begin{array}{c\|c} I & t \end{array}\right]_{2\times3}$	2	orientation	
rigid (Euclidean)	$\left[\begin{array}{c\|c} R & t \end{array}\right]_{2\times3}$	3	lengths	
similarity	$\left[\begin{array}{c\|c} sR & t \end{array}\right]_{2\times3}$	4	angles	
affine	$\left[\begin{array}{c} A \end{array}\right]_{2\times3}$	6	parallelism	
projective	$\left[\begin{array}{c} \tilde{H} \end{array}\right]_{3\times3}$	8	straight lines	

Table 2.1 Hierarchy of 2D coordinate transformations. Each transformation also preserves the properties listed in the rows below it, i.e., similarity preserves not only angles but also parallelism and straight lines. The 2×3 matrices are extended with a third $[0^T\ 1]$ row to form a full 3×3 matrix for homogeneous coordinate transformations.

Distortion Removal Part 1: Projective up to Affine

- Whereas every affine transform maps l_∞ to l_∞, a general projective transform maps l_∞ to a physical line that we call the **vanishing line**.
  - This is illustrated in the following Figure.



- The distortion in an image that results in the formation of one or more vanishing points and vanishing lines in the plane of the image is specifically projective.
  - Meaning that it is over and above the distortion introduced by affine part of the overall transformation.

Distortion Removal: Projective up to Affine

- If we apply a homography to an image that sends the vanishing line(s) back to l_∞, the remaining distortion in the image will be purely affine.
- What's interesting is that the homography for removing the projective part of the distortion can be estimated from the parameters of the vanishing line.
  - That is, if the vanishing line is $l = \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix}$, the homography that takes the vanishing line back to l_∞ is given by: $H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$.

Distortion Removal: Projective up to Affine

- To prove the assertion in the previous slide, recall that when points are transformed by H, lines are transformed by H^{-T}. So we write first the matrix for $H^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -l_1/l_3 & -l_2/l_3 & 1/l_3 \end{bmatrix}$ implying that

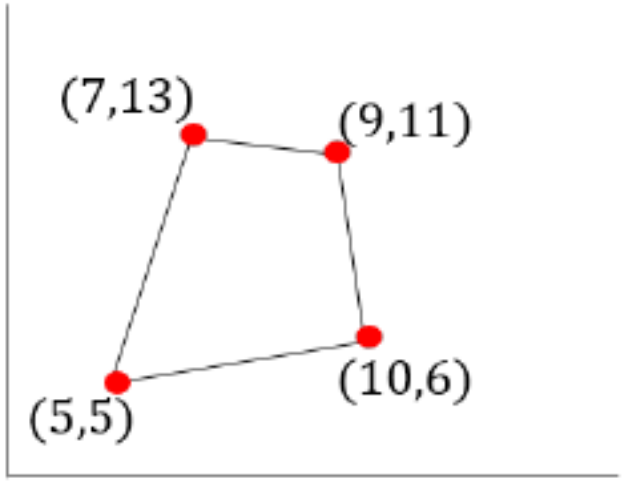
$$H^{-T} = \begin{bmatrix} 1 & 0 & -l_1/l_3 \\ 0 & 1 & -l_2/l_3 \\ 0 & 0 & 1/l_3 \end{bmatrix}.$$

- The last step of the proof consists of showing that

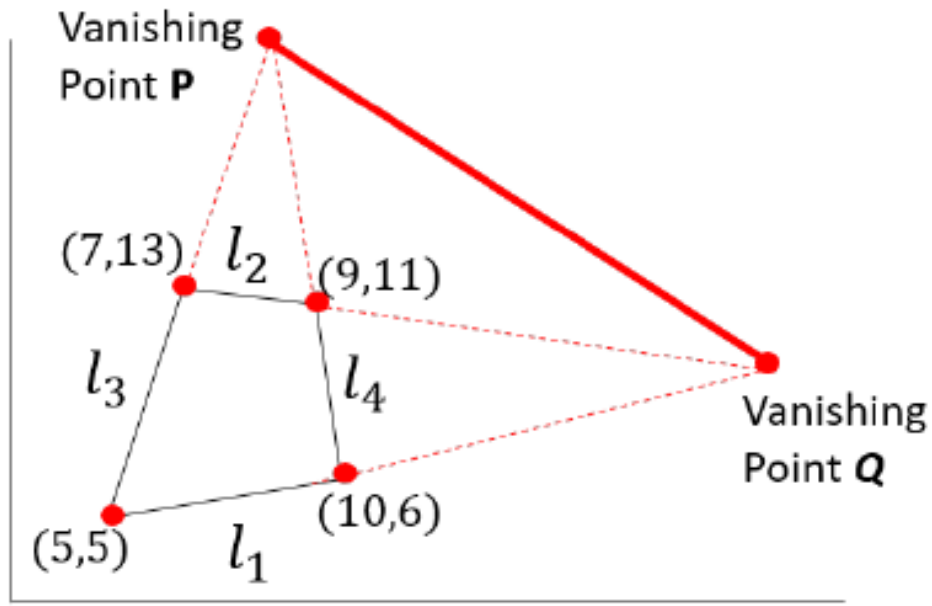
$$H^{-T} \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix} = \begin{bmatrix} 1 & 0 & -l_1/l_3 \\ 0 & 1 & -l_2/l_3 \\ 0 & 0 & 1/l_3 \end{bmatrix} \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix} = \begin{pmatrix} l_1 - l_1 \\ l_2 - l_2 \\ l_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = l_\infty.$$

Exercise

- In a photograph of a wall at which pictures are mounted very high on the wall and off to the right of the photographer, the pixel coordinates of the corners of a rectangular frame are at (5,5), (10,6), (7,13), and (9,11): The photo obviously suffers from projective distortion. What 3×3 homography will get rid of this distortion?



Solution



- $l_1 = \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} 10 \\ 6 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ -20 \end{pmatrix}$, $l_2 = \begin{pmatrix} 7 \\ 13 \\ 1 \end{pmatrix} \times \begin{pmatrix} 9 \\ 11 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -40 \end{pmatrix}$
- $l_3 = \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix} \times \begin{pmatrix} 7 \\ 13 \\ 1 \end{pmatrix} = \begin{pmatrix} -8 \\ 2 \\ 30 \end{pmatrix}$, $l_4 = \begin{pmatrix} 10 \\ 6 \\ 1 \end{pmatrix} \times \begin{pmatrix} 9 \\ 11 \\ 1 \end{pmatrix} = \begin{pmatrix} -5 \\ -1 \\ 56 \end{pmatrix}$

Solution

- $Q = l_1 \times l_2 = \begin{pmatrix} -160 \\ -80 \\ -12 \end{pmatrix}$
- $P = l_3 \times l_4 = \begin{pmatrix} 142 \\ 298 \\ 18 \end{pmatrix}$
- Vanishing Line $= P \times Q = \begin{pmatrix} l_1 \\ l_2 \\ l_3 \end{pmatrix} = \begin{pmatrix} 2136 \\ 1176 \\ -36320 \end{pmatrix}$, $H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$

VERY IMPORTANT MATERIAL:
DISTORTION PROJECTIVE REMOVAL AND IMPLEMENTING IMAGE WARPING (PYTHON/MATLAB)

Will be needed when we will talk about Image Alignment and Stitching in Chapter 8 of the textbook

Distortion Projective Removal: Part 2

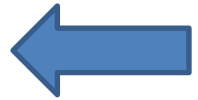
Point Correspondences for Estimating a Homography

- As you know already, given a point X in a planer scene and its corresponding pixel X' in the image plane, for most cameras we can write $X' = HX$ assuming that X and X' are expressed using the homogeneous coordinates:

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \text{ and } X' = \begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix}. \text{ With } H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}, \text{ we end up}$$

with :

- $x_1' = h_{11}x_1 + h_{12}x_2 + h_{13}x_3$
- $x_2' = h_{21}x_1 + h_{22}x_2 + h_{23}x_3$
- $x_3' = h_{31}x_1 + h_{32}x_2 + h_{33}x_3$



$$\begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Distortion Projective Removal: Part 2

Point Correspondences for Estimating a Homography

- Denoting the physical scene coordinate by (x, y) , and the physical pixel coordinates by (x', y'), we have:
  - $x = \frac{x_1}{x_3}$, $y = \frac{x_2}{x_3}$
  - $x' = \frac{x_1'}{x_3'}$, $y' = \frac{x_2'}{x_3'}$
- So, we can write for the physical coordinates of the image pixel:
  - $x' = \frac{h_{11}x_1 + h_{12}x_2 + h_{13}x_3}{h_{31}x_1 + h_{32}x_2 + h_{33}x_3} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$
  - $y' = \frac{h_{21}x_1 + h_{22}x_2 + h_{23}x_3}{h_{31}x_1 + h_{32}x_2 + h_{33}x_3} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$
- Note that by dividing through by x_3 on the left-hand sides we get purely in terms of just the physical coordinates on both sides.

Distortion Projective Removal: Part 2

Point Correspondences for Estimating a Homography

- These expressions for the physical pixel coordinates can be rewritten in the following form:
  - $h_{11}x + h_{12}y + h_{13} + \underline{h_{21}(0) + h_{22}(0) + h_{23}(0)} - h_{31}x'x - h_{32}x'y - h_{33}x' = 0$
  - $\underline{h_{11}(0) + h_{12}(0) + h_{13}(0)} + h_{21}x + h_{22}y + h_{23} - h_{31}y'x - h_{32}y'y - h_{33}y' = 0$
- Thus **a single point correspondence** between the original scene and the image gives us **two linear equations** for the elements of H.

Distortion Projective Removal: Part 2

Point Correspondences for Estimating a Homography

- **Question:** How many equations and points do we need to solve the previous equations
  - Note that each correspondence point gives us two equations. More over we have 9 unknown parameters $(h_{11}, h_{12}, ..., h_{33})$.
  - Thus, we need 4 or 5 points.
    - But, remember H is homogeneous
    - *i.e.* multiplying by any constant or altering any one element will preserve the mapping, which means we can choose any h_{ij} equal to one or any number
  - From now on, we will assume $h_{33} = 1$
    - We will explain the main reason later.
  - So, what is remaining is just 8 unknowns (8 groups or ratios).
    - $h_{11} : h_{12} : h_{13} : h_{21} : h_{22} : h_{23} : h_{31} : h_{32} : h_{33}$
    - Therefore, we only need 4 points

Distortion Projective Removal: Part 2
Point Correspondences for Estimating a Homography

- So, with 4 point correspondences, we will get eight equations for the nine unknowns of H. However, since H is homogeneous (that is, since only the ratios of the elements of H are important), we only need to calculate H within a multiplicative constant.
- This can be done with four point-to-point correspondences **provided** that the **eight equations are linearly independent**
  - A condition that is satisfied when no three of the four points fall on a single straight line (non-collinear points).

Distortion Projective Removal: Part 2

Point Correspondences for Estimating a Homography

- As such, given 4 point correspondences we can solve for H as follows:
  - Form the matrix $A_{8\times 9}\overrightarrow{h_{9\times 1}} = \overrightarrow{0_{8\times 1}}$, where
    - $\overrightarrow{h} = (h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}, h_{33})^T$
    - $A = \begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x'x & -x'y & -x' \\ 0 & 0 & 0 & x & y & 1 & -y'x & -y'y & -y' \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \end{bmatrix}_{8\times 9}$
  - Solve for $\overrightarrow{h}$ using SVD (Singular Value Decomposition)
    - if $SVD(A) = U\Sigma V^T$, then $\vec{h}$ must be the last column of the matrix V (or the basis vector of the null space of A).

Distortion Projective Removal: Part 2
Point Correspondences for Estimating a Homography and Image Warping

- Demo Using Python/Matlab:
  - Code will be posted on the BB.

